WORKED SOLUTION

STEP III 1989, Question 11

Mechanics • Oblique Impact of Spheres

STEP III 1989, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q11

The key idea

First pin down the geometry at the moment of impact: it fixes the line of centres, and everything else follows from it. Then use the standard recipe for an oblique impact of smooth spheres when one of them is constrained. The impulse between the spheres acts along the line of centres. Newton's law of restitution applies to velocity components along that line. And instead of conservation of momentum, which fails because the string or the pivot supplies an impulse of its own, we use whichever equation the constraint leaves untouched: horizontal momentum for A with the string, angular momentum about O with the rod.

Solution

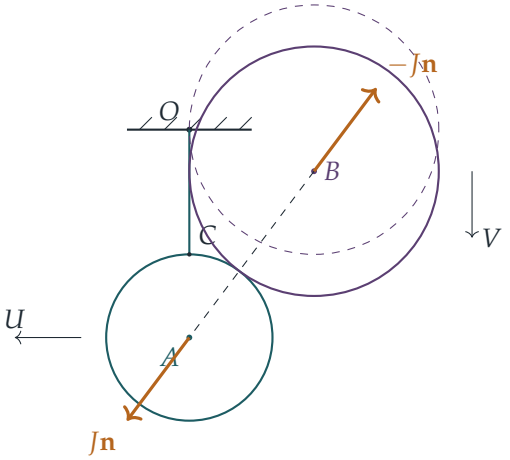
Geometry at impact

We model both spheres as smooth and uniform, and the string as light and inextensible. Before the impact the string hangs vertically. Since C lies on the line OA, we have $OA = 3a + 2a = 5a$, with A directly below O.

Sphere B starts with $OB = 3a$ horizontal and falls vertically. Let it fall a distance h. At impact the spheres touch, so $AB = 2a + 3a = 5a$. The horizontal separation of A and B is $3a$, so their vertical separation is $\sqrt{25a^2 - 9a^2} = 4a$. Hence B is $4a$ above A, or a below the level of O, and $h = a$.

So the line of centres is a 3–4–5 triangle: from B to A it goes $3a$ horizontally (towards OA) and $4a$ down. Let $\mathbf{n}$ be the unit vector from B towards A, and take the x-axis horizontal, pointing from OA towards B. Then

$$\mathbf{n} = \left(-\tfrac{3}{5}, -\tfrac{4}{5}\right).$$



The dashed sphere marks where B starts, and the solid one where it is at impact.

Impact with the string

Let J be the magnitude of the impulse between the spheres, so A receives $J\mathbf{n}$ and B receives $-J\mathbf{n}$. The spheres are smooth, so $J\mathbf{n}$ acts along the line of centres and passes through A. The string's impulse acts along OC, which also passes through A. So nothing gives sphere A any spin, and A moves as a particle on a circle of radius $5a$ about O. Immediately after the impact the string is still vertical, so A moves horizontally, with speed U in the direction of negative x.

Sphere A, horizontally. The string gives no horizontal impulse, so

$$3mU = \tfrac{3}{5}J, \qquad \text{that is,} \qquad J = 5mU.$$

(Vertically, the string supplies an upward impulse $\frac{4}{5}J$ that cancels the downward part of $J\mathbf{n}$. This is positive, so the string really does stay taut.)

Sphere B. Before the impact $\mathbf{v}_B = (0, -V)$, whose component along $\mathbf{n}$ is $\frac{4}{5}V$. The impulse $-J\mathbf{n}$ changes only this component, and it reduces it by $J/(25m)$:

$$\mathbf{v}'_B \cdot \mathbf{n} = \tfrac{4}{5}V - \frac{J}{25m} = \tfrac{4}{5}V - \tfrac{1}{5}U.$$

Ask yourself Why can't we just conserve momentum along the line of centres?

The string's impulse on A is vertical, and that has a component along $\mathbf{n}$. So the total momentum along $\mathbf{n}$ changes. The only direction in which no external impulse acts is horizontal, and that is the equation we used for A.

Restitution. Along $\mathbf{n}$, the speed of approach is $\frac{4}{5}V - 0$, since A starts at rest. The speed of separation is $\mathbf{v}'_A \cdot \mathbf{n} - \mathbf{v}'_B \cdot \mathbf{n}$, where $\mathbf{v}'_A \cdot \mathbf{n} = (-U)(-\frac{3}{5}) = \frac{3}{5}U$. So

$$\tfrac{3}{5}U - \left(\tfrac{4}{5}V - \tfrac{1}{5}U\right) = \tfrac{3}{4} \cdot \tfrac{4}{5}V \quad \Longrightarrow \quad \tfrac{4}{5}U = \tfrac{7}{5}V \quad \Longrightarrow \quad U = \tfrac{7}{4}V.$$

Impact with the rod

Now the sphere and the rod form a single rigid body turning about O. The pivot can supply an impulse in any direction, so horizontal momentum is no longer conserved. But the pivot's impulse acts at O, so we take moments about O. Its moment of inertia about O is, by the parallel axis theorem,

$$I = \tfrac{2}{5}(3m)(2a)^2 + 3m(5a)^2 = \tfrac{24}{5}ma^2 + 75ma^2 = \tfrac{399}{5}ma^2.$$

The impulse $J\mathbf{n}$ acts through A, which is a depth $5a$ below O. Only its horizontal component $\frac{3}{5}J$ has a moment about O, and that moment is $5a \cdot \frac{3}{5}J = 3aJ$. If the body's angular speed just after the impact is ω, then A moves horizontally with speed $U' = 5a\omega$, and

$$I\omega = 3aJ \quad\Longrightarrow\quad \tfrac{399}{5}ma^2 \cdot \frac{U'}{5a} = 3aJ \quad\Longrightarrow\quad J = \tfrac{133}{25}\,mU'.$$

The equations for B and for restitution are exactly as before, because they involve only the components along $\mathbf{n}$. So

$$\tfrac{3}{5}U' - \left(\tfrac{4}{5}V - \frac{J}{25m}\right) = \tfrac{3}{5}V \quad\Longrightarrow\quad \left(\tfrac{3}{5} + \tfrac{133}{625}\right)U' = \tfrac{7}{5}V \quad\Longrightarrow\quad \tfrac{508}{625}U' = \tfrac{7}{5}V,$$

so $U' = \frac{875}{508}V$. Comparing this with $U = \frac{7}{4}V$,

$$\frac{U'}{U} = \frac{875}{508} \cdot \frac{4}{7} = \frac{500}{508} = \frac{125}{127},$$

so the speed of A immediately after the impact is $125U/127$, as required.

It makes sense that the answer is a little smaller than before. With the string, A resists horizontal motion like a particle of mass $3m$. With the rod, the sphere must also start spinning, and its effective horizontal mass at A becomes $I/(5a)^2 = \frac{399}{125}m$, which is slightly more than $3m$.

Checks

- Energy: with $m = a = V = 1$, the kinetic energy is 12.5 before the impact. Afterwards it is 11.625 with the string and about 11.58 with the rod. Both are smaller, as they must be when $e < 1$.
- Elastic limit: with $e = 1$ and the string, the same steps give $\frac{4}{5}U = \frac{8}{5}V$, so $U = 2V$, and kinetic energy is conserved exactly. This confirms the signs.
- A computer solved both impacts in exact fractions, with $a = m = V = 1$. It found $U = 7/4$ and $U' = 875/508$, giving the ratio $125/127$. It also checked that B's tangential velocity is unchanged and that B falls a distance a before contact.

Common slips

- Conserving total momentum along the line of centres. The string or pivot supplies an external impulse with a component in that direction.

- Forgetting that the sphere on the rod rotates. Its moment of inertia about O needs the $\frac{2}{5}(3m)(2a)^2$ term as well as $3m(5a)^2$.
- Taking the line of centres at $45°$, or as vertical. The touching condition fixes it as a 3–4–5 triangle.

Beyond STEP

Constrained impacts are a first taste of *impulsive motion of rigid bodies*. The general rule is that an impulse at a fixed pivot drops out of the angular momentum equation about that pivot. This leads on to the idea of the *centre of percussion*: the point where you can strike a pivoted body so that the pivot feels no impulse at all, which is why a cricket bat has a "sweet spot". Kleppner and Kolenkow's *An Introduction to Mechanics* treats angular momentum and rigid-body impulses at exactly this level.

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